

Optimization in Machine Learning

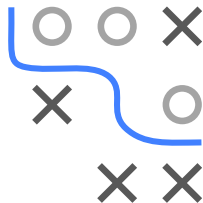
Foundations

Matrix Calculus



Learning goals

- Rules of matrix calculus
- Connection of gradient, Jacobian and Hessian

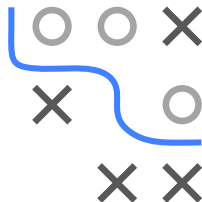


SCOPE

- $\mathcal{X}/\mathcal{Y}$ denote space of **independent/dependent** variables
- Identify dependent variable y with a **function**
 $f: \mathcal{X} \rightarrow \mathcal{Y}, x \mapsto f(x)$
- Assume y sufficiently smooth
- In matrix calculus, x and y can be **scalars**, **vectors**, or **matrices**
- We denote vectors/matrices in **bold** lowercase/uppercase letters

Type	scalar x	vector $\mathbf{x}$	matrix $\mathbf{X}$
scalar y	dy/dx	$dy/d\mathbf{x}$	$dy/d\mathbf{X}$
vector $\mathbf{y}$	$d\mathbf{y}/dx$	$d\mathbf{y}/d\mathbf{x}$	—
matrix $\mathbf{Y}$	$d\mathbf{Y}/dx$	—	—

- This notation is also referred to as *Leibniz notation*



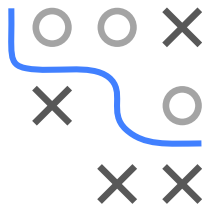
LEIBNIZ NOTATION CONVENTION

- Instead of writing $f(x)$ everywhere, we replace the function f with the variable y .
- This helps clarify relationships when multiple functions or variables are involved, especially in contexts like partial derivatives or matrix calculus.
- Also applicable to partial derivatives: For $y = f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{x} \mapsto f(\mathbf{x}) = f(x_1, x_2, \dots, x_n)$, the partial derivative of y w.r.t. x_i is dy/dx_i .

- **Examples:**

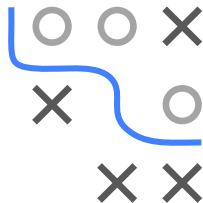
- $y = x^3 + 5x \implies \frac{dy}{dx} = 3x^2 + 5$

- $y = x_1^2 + 3x_2 \implies \frac{dy}{dx_1} = 2x_1, \quad \frac{dy}{dx_2} = 3$



DERIVATIVES OF SCALAR-VALUED FUNCTIONS

Type	scalar x	vector $\mathbf{x}$	matrix $\mathbf{X}$
scalar y	dy/dx	$dy/d\mathbf{x}$	$dy/d\mathbf{X}$
vector $\mathbf{y}$	$d\mathbf{y}/dx$	$d\mathbf{y}/d\mathbf{x}$	—
matrix $\mathbf{Y}$	$d\mathbf{Y}/dx$	—	—



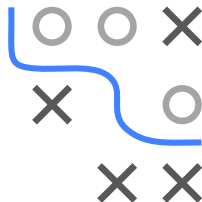
- $dy/d\mathbf{x}$ is the gradient from the previous slide deck
- When the input is a matrix the concept remains the same, i.e. for $y = f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$, $\mathbf{X} \mapsto f(\mathbf{X})$

$$\frac{dy}{d\mathbf{X}} = \begin{pmatrix} \frac{\partial f}{\partial x_{11}} & \cdots & \frac{\partial f}{\partial x_{1n}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_{m1}} & \cdots & \frac{\partial f}{\partial x_{mn}} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

DERIVATIVES: UNIVARIATE AND JACOBIAN

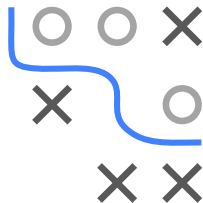
Type	scalar x	vector $\mathbf{x}$	matrix $\mathbf{X}$
scalar y	dy/dx	$dy/d\mathbf{x}$	$dy/d\mathbf{X}$
vector $\mathbf{y}$	$d\mathbf{y}/dx$	$d\mathbf{y}/d\mathbf{x}$	—
matrix $\mathbf{Y}$	$d\mathbf{Y}/dx$	—	—

- dy/dx is the univariate derivative y'
- $d\mathbf{y}/d\mathbf{x}$ is the Jacobian from the previous slide deck



DERIV. OF FUNCTIONS WITH SCALAR INPUTS

Type	scalar x	vector $\mathbf{x}$	matrix $\mathbf{X}$
scalar y	dy/dx	$dy/d\mathbf{x}$	$dy/d\mathbf{X}$
vector $\mathbf{y}$	$d\mathbf{y}/dx$	$d\mathbf{y}/d\mathbf{x}$	—
matrix $\mathbf{Y}$	$d\mathbf{Y}/dx$	—	—



- Here, for univariate $f_{ij} : \mathbb{R} \rightarrow \mathbb{R}$, $\mathbf{y}$ ($n = 1$) or $\mathbf{Y}$ ($n > 1$) are equal to a function $f : \mathbb{R} \rightarrow \mathbb{R}^{m \times n}$, $x \mapsto (f_{ij}(x))_{i=1, \dots, m; j=1, \dots, n}$ and the derivatives are, respectively given by

$$\frac{d\mathbf{y}}{dx} = \begin{pmatrix} \frac{\partial f_1}{\partial x} \\ \vdots \\ \frac{\partial f_m}{\partial x} \end{pmatrix} \in \mathbb{R}^m; \quad \frac{d\mathbf{Y}}{dx} = \begin{pmatrix} \frac{\partial f_{11}}{\partial x} & \cdots & \frac{\partial f_{1n}}{\partial x} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_{m1}}{\partial x} & \cdots & \frac{\partial f_{mn}}{\partial x} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

MULTIVARIATE DIFFERENTIATION RULES

- Basic rules from single-variable calculus still apply.
- But, for $\mathbf{x} \in \mathbb{R}^n$: gradients are vectors/matrices (order matters).

Key Rules:

- **Sum:**

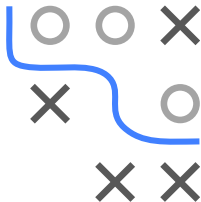
$$\frac{d}{d\mathbf{x}}(f + g) = \frac{df}{d\mathbf{x}} + \frac{dg}{d\mathbf{x}}$$

- **Product:**

$$\frac{d}{d\mathbf{x}}(fg) = \frac{df}{d\mathbf{x}} g + f \frac{dg}{d\mathbf{x}}$$

- **Chain:**

$$\frac{d}{d\mathbf{x}}((f \circ g)(\mathbf{x})) = \frac{d}{d\mathbf{x}}(f(g(\mathbf{x}))) = \frac{df}{dg} \frac{dg}{d\mathbf{x}}$$



DETAILS ON THE CHAIN RULE

- Suppose

- we have functions $\mathbf{g} : S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{f} : T \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^\ell$
- $\mathbf{a} \in S$ is a point such that $\mathbf{g}(\mathbf{a}) \in T \Rightarrow \mathbf{f} \circ \mathbf{g}(\mathbf{x}) = \mathbf{f}(\mathbf{g}(\mathbf{x}))$ is well-defined for all $\mathbf{x}$ close to $\mathbf{a}$

- Then, if $\mathbf{g}$ is differentiable at $\mathbf{a}$ and $\mathbf{f}$ is differentiable at $\mathbf{g}(\mathbf{a})$

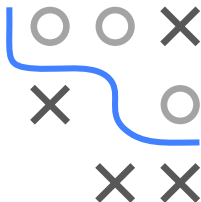
$\Rightarrow \mathbf{f} \circ \mathbf{g}$ is differentiable at $\mathbf{a}$, and the derivative $\frac{d\mathbf{f}}{d\mathbf{g}} \frac{d\mathbf{g}}{d\mathbf{x}}$, is equal to

$$\nabla_{\mathbf{a}} \mathbf{f} \circ \mathbf{g} \hat{=} \mathbf{J}_{\mathbf{f} \circ \mathbf{g}}(\mathbf{a}) = \mathbf{J}_{\mathbf{f}}(\mathbf{g}(\mathbf{a})) \mathbf{J}_{\mathbf{g}}(\mathbf{a}) \hat{=} \nabla_{\mathbf{g}(\mathbf{a})} \mathbf{f} \nabla_{\mathbf{a}} \mathbf{g} \in \mathbb{R}^{\ell \times n}$$

(See Chapter 2.3 of the UofT course *MAT237 - Multivariable Calculus* for proof.)

- We can also write $\mathbf{f}$ as a function of $\mathbf{y} = (y_1, \dots, y_m) \in \mathbb{R}^m$, and $\mathbf{g}$ as a function of $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$. Then for each $k = 1, \dots, \ell$ and $j = 1, \dots, n$

$$[\mathbf{J}_{\mathbf{f} \circ \mathbf{g}}(\mathbf{a})]_{kj} = \frac{\partial}{\partial x_j} (f_k \circ \mathbf{g})(\mathbf{a}) = \sum_{i=1}^m \frac{\partial f_k}{\partial y_i}(\mathbf{g}(\mathbf{a})) \frac{\partial g_i}{\partial x_j}(\mathbf{a})$$

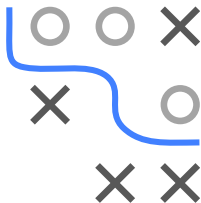


HELPFUL CALCULATION RULES

Let $\mathbf{a}$, $\mathbf{b}$ denote vectors, $\mathbf{X}$, $\mathbf{A}$ matrices, and $f(\mathbf{X})^{-1}$ the inverse of $f(\mathbf{X})$ if it exists.

- $\frac{d\mathbf{x}^T \mathbf{a}}{d\mathbf{x}} = \mathbf{a}^T, \frac{d\mathbf{a}^T \mathbf{x}}{d\mathbf{x}} = \mathbf{a}^T$
- $\frac{d\mathbf{X}\mathbf{a}}{d\mathbf{a}} = \mathbf{X}, \frac{d\mathbf{a}^T \mathbf{X}}{d\mathbf{a}} = \mathbf{X}^T$
- $\frac{d\mathbf{a}^T \mathbf{X} \mathbf{b}}{d\mathbf{X}} = \mathbf{a} \mathbf{b}^T$
- $\frac{d\mathbf{x}^T \mathbf{A} \mathbf{x}}{d\mathbf{x}} = \mathbf{x}^T (\mathbf{A} + \mathbf{A}^T)$
- For a symmetric matrix $\mathbf{W}$, $\frac{d}{ds}(\mathbf{x} - \mathbf{A} \mathbf{s})^T \mathbf{W} (\mathbf{x} - \mathbf{A} \mathbf{s}) = -2(\mathbf{x} - \mathbf{A} \mathbf{s})^T \mathbf{W} \mathbf{A}$
- $\frac{d}{d\mathbf{X}} f(\mathbf{X})^T = \left(\frac{df(\mathbf{X})}{d\mathbf{X}} \right)^T$
- $\frac{d}{d\mathbf{X}} f(\mathbf{X})^{-1} = -f(\mathbf{X})^{-1} \frac{df(\mathbf{X})}{d\mathbf{X}} f(\mathbf{X})^{-1}$
- $\frac{d\mathbf{a}^T \mathbf{X}^{-1} \mathbf{b}}{d\mathbf{X}} = -(\mathbf{X}^{-1})^T \mathbf{a} \mathbf{b}^T (\mathbf{X}^{-1})^T$

Note: to compute gradients of matrices with respect to vectors (or other matrices) we need *tensors*, see chapter 5.4 of Deisenroth for more.



EXAMPLE: LOGISTIC REGRESSION I

- Let's say, for data in $\mathbb{R}^{n \times m}$ we're trying to minimize the risk in logistic regression by finding the gradient for negative log loss:

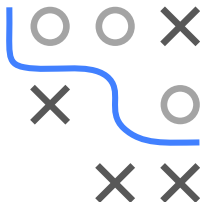
$$-\ell(\theta) = \sum_{i=1}^n -y^{(i)} \log \left(\pi \left(\mathbf{x}^{(i)} \mid \theta \right) \right) - \left(1 - y^{(i)} \right) \log \left(1 - \pi \left(\mathbf{x}^{(i)} \mid \theta \right) \right)$$

where $\pi(\mathbf{x} \mid \theta) = s(f(\theta, \mathbf{x}))$ with $f(\theta, \mathbf{x}) = \theta^T \mathbf{x}$ and $s(x) = \frac{1}{1 + \exp(-x)}$.

⇒ We want to find

$$\nabla_{\theta} - \ell(\theta) = -\nabla_{\theta} \ell(\theta) = - \sum_{i=1}^n \underbrace{y^{(i)} \log(s(f(\theta, \mathbf{x}^{(i)}))) + (1 - y^{(i)}) \log(1 - s(f(\theta, \mathbf{x}^{(i)})))}_{=: y_i \log(s_i) + (1 - y_i) \log(1 - s_i)}$$

- $s_i = s(f_i)$; $h_i := -[y_i \log(s_i) + (1 - y_i) \log(1 - s_i)]$; $f_i := \theta^T \mathbf{x}^{(i)}$



EXAMPLE: LOGISTIC REGRESSION II

- We can now directly apply the chain rule:

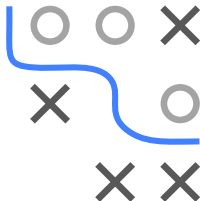
$$-\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n \nabla_{\theta} h_i \circ s_i \circ f_i = \sum_{i=1}^n \frac{dh_i}{ds_i} \frac{ds_i}{df_i} \frac{df_i}{d\theta}$$

- Given that, $\forall i \in \{1, \dots, n\}$

$$\frac{dh_i}{ds_i} = \frac{1 - y_i}{1 - s_i} - \frac{y_i}{s_i}; \quad \frac{ds_i}{df_i} = s(f_i)(1 - s_i(f_i)); \quad \frac{df_i}{d\theta} = (\mathbf{x}^{(i)})^T$$

- we get that $-\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n \frac{dh_i}{ds_i} \frac{ds_i}{df_i} \frac{df_i}{d\theta}$ equals

$$\begin{aligned} & \sum_{i=1}^n \left[\frac{1 - y^{(i)}}{1 - s(f(\theta, \mathbf{x}^{(i)}))} - \frac{y^{(i)}}{s(f(\theta, \mathbf{x}^{(i)}))} \right] \cdot \left[s(f(\theta, \mathbf{x}^{(i)}))(1 - s(f(\theta, \mathbf{x}^{(i)}))) \right] \left(\mathbf{x}^{(i)} \right)^T \\ &= \sum_{i=1}^n \left(s(f(\theta, \mathbf{x}^{(i)})) - y^{(i)} \right) \left(\mathbf{x}^{(i)} \right)^T \in \mathbb{R}^{1 \times m} \end{aligned}$$



EX: LOG. REGR. – ALTERNATIVE NOTATION

This example highlights how using Leibniz notation can make things easier.

Alternatively, we could write this example out as follows:

- Given the sum rule, it suffices to find the derivative of

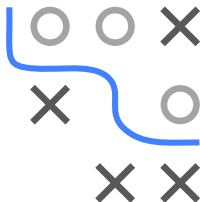
$$-[y \log(s(f(\theta, \mathbf{x}))) + (1 - y) \log(1 - s(f(\theta, \mathbf{x})))]$$

- Define $h(y, z) := -[y \log(z) + (1 - y) \log(1 - z)]$.
- Now, what we are looking for is

$$\nabla_{\theta} h(y, \cdot) \circ s \circ f(\cdot, \mathbf{x}).$$

where f is a function from $\mathbb{R}^p$ to $\mathbb{R}$ and both h and s are functions from $\mathbb{R}$ to $\mathbb{R} \implies$ by the chain rule

$$\nabla_{\theta} h(y, \cdot) \circ s \circ f(\cdot, \mathbf{x}) = \frac{dh}{ds} \frac{ds}{df} \frac{df}{d\theta} \in \mathbb{R}^{1 \times p}.$$



EX: LOG. REGR. – ALTERNATIVE NOTATION

- Using the fact that

$$\frac{dh}{dz} = \frac{1-y}{1-z} - \frac{y}{z}; \quad \frac{ds}{dx} = s(x)(1-s(x)); \quad \frac{df}{d\theta} = \mathbf{x}$$

we get

$$\begin{aligned} [\nabla_{\theta} h(y, \cdot) \circ s \circ f(\cdot, \mathbf{x})]_j &= \left[\frac{1-y}{1-s(f(\theta, \mathbf{x}))} - \frac{y}{s(f(\theta, \mathbf{x}))} \right] \cdot [s(f(\theta, \mathbf{x}))(1-s(f(\theta, \mathbf{x})))] \cdot \frac{\partial f}{\partial \theta_j} \\ &= (s(f(\theta, \mathbf{x})) - y) \mathbf{x}_j \end{aligned}$$

$$\implies \nabla_{\theta} h(y, \cdot) \circ s \circ f(\cdot, \mathbf{x}) = (s(f(\theta, \mathbf{x})) - y) \mathbf{x}^T$$

- Plugging this back into the original sum term we get

$$-\nabla_{\theta} \ell(\theta) = \sum_{i=1}^n \left(s \left(f \left(\theta, \mathbf{x}^{(i)} \right) \right) - y^{(i)} \right) \left(\mathbf{x}^{(i)} \right)^T \in \mathbb{R}^{1 \times m},$$

the same as before.

